MISCELLANEOUS NOTES IN PHYSICS

And other stuff

Author
Aditya Chugh
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a.chugh@mail.utoronto.ca

Contents

1 Different Approaches to Quantum Dynamics **3**

1.1 Schrödinger Picture (SP) 3

1.2 Heisenberg Picture (HP) 3

2 Perturbation theory **4**

2.1 Time-independent, non-degenerate 4

2.2 Time-independent, degenerate 5

2.3 Time-dependent 7

2.3.1 Interaction Picture (IP) 8

2.3.2 Perturbative solution 9

2.3.3 Transition amplitudes 10

2.3.4 Types of Perturbations and Fermi's Golden Rule 11

3 Relativistic formulation of Electrodynamics **12**

4 The Electromagnetic field tensor and a new way of looking at E&M **15**

5 Density Matrices **18**

6 Generators of transformations **18**

7 Transverse and Longitudinal components of fields **18**

8 Plasmas **18**

9 Retarded potentials and Maxwell's stress tensor **18**

10 Contravariance and Covariance **18**

11 Dispersion relations **18**

12 Redshifts and time in Cosmology **18**

13 Classical Field Theory **18**

1 Different Approaches to Quantum Dynamics

1.1 Schrödinger Picture (SP)

1.2 Heisenberg Picture (HP)

2 Perturbation theory

Say you're studying a quantum system that's similar to one you've studied before. The Hamiltonian of this new system doesn't differ much from that of the older system - just an additional new term that's small, in some sense, in its effects [1]. You can imagine introducing this perturbation with a strength factor - although the form of the perturbation is fixed (maybe you're turning on an external magnetic field or introducing a new potential function V), this strength factor parameterizes the magnitude of your introduced change. That is, a number λ that bridges the gap between no perturbation (for $\lambda = 0$) and complete perturbation (for $\lambda = 1$).

Then, the essence of perturbation theory is that small changes in your Hamiltonian should introduce small changes in your system in a continuous way (of which we keep track through λ). If your original Hamiltonian had some eigenstates $|n\rangle$ with eigenvalues E_n, then the new eigenstates should be close to the original ones (this can be made precise using norms since we're in a Hilbert space) and the energy values should also shift up or down in a continuous manner by small amounts (for small enough perturbations, that is).

There are three situations we will consider here. The first two concern time-independent Hamiltonians, but with non-degenerate and degenerate energy levels respectively. For the final situation, we look at time-dependent Hamiltonians.

2.1 Time-independent, non-degenerate

We assume that no correction (of any order) has a component along $\left\langle n^{(0)}\right|$ [2]. Wikipedia calls this an intermediate normalization (why so, I am yet to find out).

We can calculate the l-th order energy correction to be,

$$E_n^{(l)} = \left\langle n^{(0)} \middle| \delta H \middle| n^{(l-1)} \right\rangle$$

A consequence of this fact is that the first order energy correction after a perturbation δH of the Hamiltonian is precisely the expectation value of this perturbation on the unperturbed state $\left|n^{(0)}\right\rangle$. That is,

$$E_n^{(1)} = \left\langle n^{(0)} \middle| \delta H \middle| n^{(0)} \right\rangle = \langle \delta H \rangle$$

We can also calculate the l-th order state correction to be,

$$\left|n^{(l)}\right\rangle = \sum_{k \neq n}^{\infty} \left|k^{(0)}\right\rangle \left\langle k^{(0)} \middle| n^{(l)} \right\rangle \tag{1}$$

$$\left|n^{(l)}\right\rangle = \sum_{k \neq n} \left|k^{(0)}\right\rangle \left[-\frac{\left\langle k^{(0)} \middle| \delta H \middle| n^{(l-1)} \right\rangle}{(E_k^{(0)} - E_n^{(0)})} + \frac{1}{(E_k^{(0)} - E_n^{(0)})} \sum_{j=1}^{l} \left\langle n^{(0)} \middle| \delta H \middle| n^{(j-1)} \right\rangle \left\langle k^{(0)} \middle| n^{(l-j)} \right\rangle \right] \tag{2}$$

[1] I'll get to what I mean by this in a bit.

[2] For a proof, confer Chapter 1 of Barton Zweibach's QM - III notes

This approach works only when we have non-degenerate (unperturbed) energy states - if there's a degenerate energy state ($k \neq n$ but $E_k^{(0)} = E_n^{(0)}$), the denominator for at least one term (in the outermost sum over k) blows up and we can't express the state correction as a well-defined sum.

Now, we discuss what it means for the perturbation to be "small", such that our assumption of being able to expand out the new energy eigenstates and eigenvalues as a power series in the parameter λ is valid. A way to quantify the size of the perturbation would be the norm of the operator δH. Now, what limit do we place on this number (which we can theoretically calculate) so that our approximation is reliable? One possible idea would be that the norm should be smaller than the entries/norm or energy eigenvalues of the original Hamiltonian. However, it turns out that it is not the absolute magnitude of the energies (of the original Hamiltonian) that ascertains the validity of this perturbative expansion, but the difference between the energies.

If we're considering the perturbation of the n-th energy level, it matters how far the adjacent energy levels lie. If they're closer together, we would (at least intuitively) expect more mixing from them, giving us bigger energy and state corrections. Since degeneracy becomes a problem, it would also make sense that it is the difference in energies that dictates the shifts of levels after perturbation (because then energy differences are equal to 0, and there may not be any non-zero interval for the λ parameter where the expansion formally works).

We also get to see another interesting result, that makes for our first sidebar.

Sidebar 2.1. Asymptotic expansions (Eg - anharmonic oscillator with quartic perturbation)

At this stage, you might be wondering if the energy levels of our system cross each other at some stage of the perturbation. Since we can make a one to one association with the unperturbed and perturbed energy states, is it ever the case that the hierarchy of states gets flipped and a lower level jumps above an adjacent upper level? This seems to be a complicated issue which I'd like to fully understand (hopefully) soon, but there is something I'd like to add here.

Sidebar 2.2. Level repulsion

2.2 Time-independent, degenerate

As stated before, the approach adopted to determine the new eigenstates and energy eigenvalues of the perturbed Hamiltonian won't really work. From a mathematical viewpoint, the issue arises due to the denominator of the sum in Eq. 2 from Section 1.1. If there is a degeneracy, the sum would blow up and the correction would be ill-defined.

Let the perturbed Hamiltonian be,

$$H(\lambda) = H^{(0)} + \lambda \delta H \tag{3}$$

In the basis of its eigenstates, let H be the diagonal matrix

$$H^{(0)} = \mathrm{diag}\{E_1^{(0)}, E_2^{(0)}, \dots, E_n^{(0)}, E_n^{(0)}, \dots, E_n^{(0)}, E_{n+1}^{(0)}, \dots\} \tag{4}$$

where $E_n^{(0)}$ repeats N times. This means $H^{(0)}$ has a degenerate N dimensional subspace V_N for eigenvalue $E_n^{(0)}$. Our analysis assumes that this is the only energy level that exhibits degeneracy.

Choose an orthonormal basis for this subspace: $\left|n^{(0)}; 1\right\rangle, \left|n^{(0)}; 2\right\rangle, \dots, \left|n^{(0)}; N\right\rangle$.

The success of this method depends on using the right basis for V_N. These basis states are the ones that evolve continuously with the application of the perturbation. Otherwise, the wrong basis states (well, wrong in this context) would seem to discontinuously jump to the appropriate basis states as soon as λ is greater than 0. This discontinuity obviously disagrees with the smooth transition that perturbation theory posits.

We write the Hilbert space as a direct sum of V_N and the subspace V of states not in V_N:

$$\mathcal{H} = V_N \oplus V \tag{5}$$

Also choose an orthonormal basis for the subspace V, namely, $\left|k^{(0)}\right\rangle$ for $k \neq n$. The appropriate orthonormality inner product relations follow.

Pick $k \in \{1, \dots, N\}$. Assume that after the application of the perturbation, the state and its energy evolves as,

$$|n; k\rangle_\lambda = \left|n^{(0)}; k\right\rangle + \lambda \left|n^{(1)}; k\right\rangle + \lambda^2 \left|n^{(2)}; k\right\rangle + \mathcal{O}(\lambda^3) \tag{6}$$

$$E_{n,k}(\lambda) = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \mathcal{O}(\lambda^3) \tag{7}$$

We assert that the perturbed state satisfies,

$$H(\lambda) |n; k\rangle_\lambda = E_{n,k}(\lambda) |n; k\rangle_\lambda \tag{8}$$

This helps us get vector equations for different orders of λ. Applying bras for different basis states (from V_N and V) allows us to extract information about corrections energy and overall states.

Assuming that the degeneracy is lifted at first order - that is, all degenerate states which had energy $E_n^{(0)}$ under the unperturbed Hamiltonian split into distinct energy levels after

perturbation. Then,

$$E_{n,k}(\lambda) = E_n^{(0)} + \lambda \delta H_{nk,nk} - \lambda^2 \sum_{p \neq n} \frac{|\delta H_{p,nk}|^2}{E_p^{(0)} - E_n^{(0)}} + \mathcal{O}(\lambda^3) \tag{9}$$

$$|n;k\rangle_\lambda = \left|n^{(0)};k\right\rangle - \lambda \left(\sum_{p \neq n} \frac{\delta H_{p,nk}}{E_p^{(0)} - E_n^{(0)}} \left|p^{(0)}\right\rangle + \sum_{l \neq k}^{l \leq N} \frac{\left|n^{(0)};l\right\rangle}{E_{n,k}^{(1)} - E_{n,l}^{(1)}} \sum_{p \neq n} \frac{\delta H_{nl,p} \delta H_{p,nk}}{E_p^{(0)} - E_n^{(0)}} \right) + \mathcal{O}(\lambda^2) \tag{10}$$

Of course, it will not always be the case that all initially degenerate levels perfectly split into distinct ones when $\lambda > 0$ (this is only when the first order energy corrections are distinct for all $\left|n^{(0)};k\right\rangle$). What may happen is that some levels split away from others, while a subset keeps to be degenerate at the first order, perhaps splitting into distinct levels at a higher order.

This means we will not be able to figure out the appropriate basis for V_N at the first order, and would need to do more work (up to at least the second order) to determine the optimal basis and thus split the levels. We now show what this would look like. For simplicity however, we consider the case where no splitting occurs at the first order, but all states separate at second order.

< insert stuff about lifting degeneracy at second order here >

2.3 Time-dependent

The approach employed for time-dependent perturbation theory is quite different from that for the time-independent case. In fact, the latter rests on a critical assumption that there exists a fixed stationary energy states for our system (that solve the Schrödinger equation), but this isn't necessarily true in the present situation.

The reason is that the existence of energy eigenstates followed form the assumption that the wave function has separable spatial and temporal parts. This is valid for a fixed Hamiltonian but not so much for a time-dependent Hamiltonian.

Our initial objective was to find these energy eigenstates because they form a basis for the state space. This is useful to understand time evolution of any configuration of our system - we could express the current state as a linear combination of these basis states, study their time evolution separately, and the linearity of the SE would allow us to connect all separate terms together to get the time evolved state.

But since there's no guarantee that we will have such eigenstates in the time-dependent H case, we might as well stick to just solving the SE directly. So yes, we're going to have to do this the more direct way. All is not lost however - we can still set up useful machinery via this method. It will have to do with transition probabilities - the probability of finding our system in a particular state after time-evolution under the perturbation (assuming it started from a specific state). This is actually pretty useful, since this is related to ideas

behind quantum mechanical scattering.

Before we start, perhaps it will be beneficial to understand what the interaction picture is.

2.3.1 Interaction Picture (IP)

I've previously described the Schrödinger picture and Heisenberg picture briefly. The interaction picture lies somewhere in the middle.

When your Hamiltonian has explicit time dependence, this interpretation helps nicely split that dependence between the operators and states. Assume that your Hamiltonian can be divided into two parts, one fixed and the other evolving in time.

$$H = H^{(0)} + \delta H(t) \tag{11}$$

So, we're treating the time-dependent part as our perturbation (the constraint is that this time dependence has to, again, be small in some sense).

The clean divide between time dependence in the interaction picture is achieved by attributing to the fixed Hamiltonian the Heisenberg picture, and to the time-dependent perturbation an evolving state under the Schrödinger picture. It's almost as if your fixed Hamiltonian is evolving in time in the background, while in the foreground the SE captures the evolution of the state with the (slightly modified) perturbation. In the end, we stick all these moving parts together and reap the profits of this elegant picture [3].

With the fixed Hamiltonian, we can define the time evolution-like operator $\mathcal{U} = e^{-iH^{(0)}t/\hbar}$. We also define a new state,

$$\left|\widetilde{\Psi}(x,t)\right\rangle = \mathcal{U}^\dagger \left|\Psi(x,t)\right\rangle \tag{12}$$

Then this state satisfies the SE,

$$i\hbar\frac{\partial}{\partial t}\left|\widetilde{\Psi}(x,t)\right\rangle = \widetilde{\delta H}(t)\left|\widetilde{\Psi}(x,t)\right\rangle \tag{13}$$

where,

$$\widetilde{\delta H}(t) = \mathcal{U}^\dagger \delta H(t)\mathcal{U} \tag{14}$$

Now, a slick formula to compute expressions of the form $\mathcal{U}^\dagger H\mathcal{U}$.

Sidebar 2.3. The Baker–Campbell–Hausdorff formula

Since the fixed Hamiltonian $H^{(0)}$ does have energy eigenstates that form a basis for our

[3] Note that this is called the interaction "picture" because it's all that is - a way to think about the math, which presumably is equivalent.

Hilbert space, we can stipulate,

$$\left|\widetilde{\Psi}(x,t)\right\rangle = \sum_n c_n(t)\left|n\right\rangle \tag{15}$$

Using this expression in Eq. 13, we can derive the following system of first order ODEs for the time-dependent coefficients of the basis states,

$$i\hbar\dot{c}_m(t) = \sum_n \widetilde{\delta H}_{mn}(t)c_n(t) \tag{16}$$

where $\widetilde{\delta H}_{mn}(t) = \langle m|\,\widetilde{\delta H}(t)\,|n\rangle$. Calling $\frac{E_m - E_n}{\hbar} = \omega_{mn}$, we can equivalently say,

$$\widetilde{\delta H}_{mn}(t) = e^{i\omega_{mn}t}\delta H_{mn}(t) \tag{17}$$

which implies,

$$i\hbar\dot{c}_m(t) = \sum_n e^{i\omega_{mn}t}\delta H_{mn}(t)c_n(t) \tag{18}$$

Why is this useful? Well, solving this (infinite) system of coupled differential equations would give us the time varying coefficients of the interaction picture wave function - a solution of the SE, by construction. And with how we've defined this state in Eq. 12,

$$\left|\Psi(x,t)\right\rangle = \mathcal{U}\left|\widetilde{\Psi}(x,t)\right\rangle = e^{-\frac{iH^{(0)}t}{\hbar}}\left|\widetilde{\Psi}(x,t)\right\rangle = \sum_n c_n(t)e^{-\frac{iE_nt}{\hbar}}\left|n\right\rangle \tag{19}$$

we've figured out how our state evolves over time under the effects of the fixed Hamiltonian and the time dependent perturbation (in theory, at least).

2.3.2 Perturbative solution

In practice, solving the system of ODEs for the coefficients of the IP state can be difficult. Now that we've developed a robust method of approximating a solution, we can go ahead and solve the interaction picture SE.

Setting up the assumed form of the solution,

$$\left|\widetilde{\Psi}(x,t)\right\rangle = \left|\widetilde{\Psi}^{(0)}(x,t)\right\rangle + \lambda\left|\widetilde{\Psi}^{(1)}(x,t)\right\rangle + \lambda^2\left|\widetilde{\Psi}^{(2)}(x,t)\right\rangle + \mathcal{O}(\lambda^3) \tag{20}$$

Inserting this into the IP Schrödinger equation, we again a vector equation with terms appearing alongside different powers of λ. Setting all these equal to 0, we get a system of equations as before.

The corrections are,

$$\left|\widetilde{\Psi}^{(0)}(x,t)\right\rangle = |\Psi(x,0)\rangle \tag{21}$$

$$\left|\widetilde{\Psi}^{(1)}(x,t)\right\rangle = \int_0^t \frac{\widetilde{\delta H}(t')}{i\hbar}\,|\Psi(x,0)\rangle\ dt' \tag{22}$$

$$\left|\widetilde{\Psi}^{(2)}(x,t)\right\rangle = \int_0^t dt'\ \frac{\widetilde{\delta H}(t')}{i\hbar}\int_0^{t'} dt''\ \frac{\widetilde{\delta H}(t'')}{i\hbar}\,|\Psi(x,0)\rangle \tag{23}$$

More generally, the n-th order equation takes the form,

$$i\hbar\partial_t\left|\widetilde{\Psi}^{(n)}(x,t)\right\rangle = \widetilde{\delta H}\left|\widetilde{\Psi}^{(n-1)}(x,t)\right\rangle \tag{24}$$

which means the solution can be written in the form,

$$\left|\widetilde{\Psi}^{(n)}(x,t)\right\rangle = \int_0^t \frac{\widetilde{\delta H}(t')}{i\hbar}\left|\widetilde{\Psi}^{(n-1)}(x,t)\right\rangle\ dt' \tag{25}$$

and the last integral can be expanded as an n times iterated integral based on the form of the (n-1)th order solution. In the end, we can apply the $H^{(0)}$ time evolution operator to get back the state $\Psi(x,t)$ we were originally working with.

2.3.3 Transition amplitudes

A direct use of the calculations we've performed previously is to figure out the probability of detecting the system in a specific state at time t, knowing the initial state we started with at $t=0$. For this purpose, the IP state ket can be used directly.

Let $|\Psi(x,0)\rangle = |n\rangle$ evolve under the time dependent Hamiltonian $H(t)$. The probability for it to be in state $|m\rangle$ at time t is,

$$P_{m\leftarrow n}(t) = |\,\langle m|\Psi(x,t)\rangle\,|^2 = \left|\,\langle m|\,e^{-\frac{iH^{(0)}t}{\hbar}}\left|\widetilde{\Psi}(x,t)\right\rangle\right|^2 = \left|\left\langle m\middle|\widetilde{\Psi}(x,t)\right\rangle\right|^2 \tag{26}$$

Since the solution for the IP state ket is a series, we can calculate the probability of transition up to an arbitrarily high order. The first term in this series is $|\Psi(x,0)\rangle = |n\rangle$, the inner product of which with $|m\rangle$ vanishes. So,

$$P_{m\leftarrow n}(t) = \left|\left\langle m\middle|\widetilde{\Psi}^{(1)}(x,t)\right\rangle + \left\langle m\middle|\widetilde{\Psi}^{(2)}(x,t)\right\rangle + \dots\right|^2 \tag{27}$$

To first order,

$$P^{(1)}_{m\leftarrow n}(t) = \left|\int_0^t \frac{\langle m|\,\widetilde{\delta H}(t')\,|\Psi(x,0)\rangle}{i\hbar}\ dt'\right|^2 = \left|\int_0^t \frac{\langle m|\,\widetilde{\delta H}(t')\,|n\rangle}{i\hbar}\ dt'\right|^2 \tag{28}$$

Then using Eq. 17,

$$P^{(1)}_{m\leftarrow n}(t) = \left|\int_0^t e^{i\omega_{mn}t'}\,\frac{\delta H_{mn}(t')}{i\hbar}\ dt'\right|^2 \tag{29}$$

A wonderful corollary of this result is,

Sidebar 2.4. $P^{(1)}_{m\leftarrow n}(t) = P^{(1)}_{n\leftarrow m}(t)$
The reason behind this is

We can also derive the corrections in terms of the coefficients $c_n(t)$ for the IP state ket. This would require expressing each correction state ket as a linear combination of the basis states of $H^{(0)}$ and forming a series in λ for the coefficient associated to say, state n.

The first order correction takes the form,

$$c^{(1)}_m(t) = \sum_n \int_0^t dt' e^{i\omega_{mn}t'} \frac{\delta H_{mn}(t')}{i\hbar} c_n(0) \tag{30}$$

and we know $c_n(0)$ for all n since we know $|\Psi(x,0)\rangle$.

By definition,

$$P_{m\leftarrow n}(t) = \left| \left\langle m \middle| \widetilde{\Psi}(x,t) \right\rangle \right|^2 = |c_m(t)|^2 \tag{31}$$

and up to first order in perturbation theory,

$$P_{m\leftarrow n}(t) \approx |c^{(0)}_m(t) + c^{(1)}_m(t)|^2 = |c_m(0) + c^{(1)}_m(t)|^2 \tag{32}$$

$$= |c_m(0)|^2 + c_m(0)^* c^{(1)}_m(t) + c^{(1)}_m(t)^* c_m(0) + \mathcal{O}(\delta H^2) \tag{33}$$

2.3.4 Types of Perturbations and Fermi's Golden Rule

(Constant and harmonic perturbations)

3 Relativistic formulation of Electrodynamics

Our notation is as follows - we will use a 3-vector, denoted by $\vec{v}$, for spatial coordinates and 4-vectors for space-time (contravariant, for now) coordinates, denoted by A^μ where $\mu = 0, 1, 2, 3$. For instance, the 4-momentum p^μ of a particle will have the components $p^0 = E/c, p^1 = p_x, p^2 = p_y, p^3 = p_z$.

Let $\vec{u} = \frac{d\vec{l}}{dt}$ denote the ordinary velocity of a particle (from a far away, external frame). So $d\vec{l}$ and dt are measured in that frame. Then the proper velocity of a particle is defined as,

$$\vec{\eta} = \frac{d\vec{l}}{d\tau} \tag{34}$$

Particularly,

$$\eta^0 = \frac{dx^0}{d\tau} = c \cdot \gamma = \frac{c}{\sqrt{1 - \frac{u^2}{c^2}}} \tag{35}$$

If the Lorentz transformation Λ takes us from frame S to frame $\bar{S}$, then the proper velocities in those two frames are related by,

$$\bar{\eta}^\mu = \Lambda^\mu_\nu \eta^\nu \tag{36}$$

This is an example of the transformation rule for 4-vector under Lorentz transforms (and in the way a 3-vector is defined by how it changes under spatial transformations, we can take this to be the defining rule for 4-vectors).

So, $\bar{\eta}^0 = \gamma(\eta^0 - \beta\eta^1)$ and so on for the transformation (which is an x-boost by speed $v = \beta c$),

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{37}$$

The transformation rules for ordinary velocities are a bit more complicated since we have to change both the infinitesimal displacement and infinitesimal time when we shift to a moving frame. They are (for a frame moving with speed v in the positive x-axis direction),

$$\bar{u}_x = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} \tag{38}$$

$$\bar{u}_y = \frac{u_y}{\gamma\left(1 - \frac{u_x v}{c^2}\right)} \tag{39}$$

$$\bar{u}_x = \frac{u_z}{\gamma\left(1 - \frac{u_x v}{c^2}\right)} \tag{40}$$

The relativistic momentum $\vec{p}$ is defined using proper velocity since if we use ordinary ve-

locities, the law of conservation of momentum will be inconsistent with the principles of relativity.

Fact check 3.1. What if we defined relativistic momentum using ordinary velocities and not proper velocities?

So [4],

$$\vec{p} = m\vec{\eta} = \gamma m\vec{u} \tag{41}$$

The four momentum is constructed using the relativistic momentum for the last three coordinates, while the zero-th component of p^μ is,

$$p^\mu = m\eta^0 = \gamma mc \tag{42}$$

Identifying $E = p^0 c$ as the relativistic energy,

$$E = \gamma mc^2 \tag{43}$$

The rest energy (put $u = 0$ so that $\gamma = 1$) is $E_{rest} = mc^2$. Then the kinetic energy is $E_{KE} = E - E_{rest}$. If $u << c$, we can Taylor expand gamma (as a function of u^2/c^2, around 0) to get,

$$E_{KE} = \frac{1}{2}mu^2 + \frac{3}{8}\frac{mu^4}{c^2} + \dots \tag{44}$$

which agrees with the classical KE up to first order. This reaffirms our faith that SR should contain in it the theory of classical kinematics in the slow velocities limit.

The transformation rules for electric and magnetic fields (under an x-boost of speed $+v$) are,

$$\begin{aligned} \bar{E}_x &= E_x & \bar{B}_x &= B_x \\ \bar{E}_y &= \gamma(E_y - vB_z) & \bar{B}_y &= \gamma\Big(B_y + \frac{v}{c^2}E_z\Big) \\ \bar{E}_z &= \gamma(E_z + vB_y) & \bar{B}_z &= \gamma\Big(B_z - \frac{v}{c^2}E_y\Big) \end{aligned} \tag{45}$$

Griffiths uses an example of a parallel plate capacitor to derive the formulae for fields perpendicular to direction of motion (x direction). The same setup is used to get the rule for E_x, while to get the rule for B_x (which is absent if you orient capacitor in y-z plane), a long solenoid aligned parallel to the x-axis is used.

While the derivation of these rules using specific setups is illustrative, especially when you haven't seen it before, you do have to make the leap of faith that they generalize to any configuration of charges/currents and thus arbitrary E and B fields. It would be nice to add a more general derivation here.

[4] Here, $\gamma = 1/\sqrt{1 - u^2/c^2}$

The electric and magnetic fields at a point in the new inertial frame are related in a simple manner for some special cases. If $\vec{B} = 0$ at a point in the original frame, then in the new frame (at that point),

$$\bar{\vec{B}} = -\frac{1}{c^2}(\vec{v} \times \bar{\vec{E}}) \tag{46}$$

Similarly, if $\vec{E} = 0$ at a point in the original frame, then in the new frame (at that point),

$$\bar{\vec{E}} = \vec{v} \times \bar{\vec{B}} \tag{47}$$

4 The Electromagnetic field tensor and a new way of looking at E&M

A very simple example of a moving point charge getting attracted to/repelled by a current carrying wire tells us that electric fields and magnetic fields are in fact, entwined with each other. One person's E-field can be another's B-field, something that has been established through many more such experiments.

We, as humans, crave elegance in our theories. We eventually do have to let go of this aspect as a requirement, since truth and accuracy take precedence in determining the future of a theory (as they should) and whether it should be pursue further. Now, with all this symmetry and intermingling, it would be nice (if not immensely satisfying) if E-fields and B-fields descended from one thing - something much more general - like a common ancestor. Thankfully, that is the case.

The fact that a Lorentz transformation mixes the components of E and B fields together eliminates the possibility of them being individual 4-vectors. Turns out, the object combining these two fields is an asymmetric, second-rank tensor.

As a definition for tensors, we can again impose certain rules on how they should behave. However, two Lorentz transformations are required to carry out this modification, instead of one as in the case of a 4-vector. Also, we can represent tensors as 2×2 matrices, thus labelling their components by the 2 indices μ, ν [5]. So, a 2-tensor $t^{\mu\nu}$ is an object that follows the following transformation rule,

$$\bar{t}^{\mu\nu} = \Lambda^{\mu}_{\lambda}\Lambda^{\nu}_{\sigma}t^{\mu\nu} \tag{48}$$

A symmetric 2-tensor would thus have 10 independent components, while an anti-symmetric 2-tensor would have 6. Starting with a general anti-symmetric 2-tensor,

$$t^{\mu\nu} = \begin{pmatrix} 0 & t_{01} & t_{02} & t_{03} \\ -t_{01} & 0 & t_{12} & t_{13} \\ -t_{02} & -t_{12} & 0 & t_{23} \\ -t_{03} & -t_{13} & -t_{23} & 0 \end{pmatrix}$$

and then applying an x-boost, we can obtain the necessary transformation rules. Directly comparing them with those for the E and B fields found before, we construct the Electromagnetic Field Tensor,

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix} \tag{49}$$

An alternate comparison (switching what transformation rule of E/B fields is compared with

[5] I haven't yet explored the math behind this, so maybe that's a digression for another day

the which general tensor rule) yields a similar object, $G^{\mu\nu}$ - the dual tensor [6]. Both these tensors give rise to the same rules for E and B fields.

We can define a current density 4-vector J^μ as,

$$J^\mu = (c\rho, \vec{J}) \tag{50}$$

where ρ is the charge density and $\vec{J}$ is the usual 3-D current density. Similarly, define the 4-vector potential A^μ as,

$$A^\mu = (V/c, \vec{A}) \tag{51}$$

where V is the scalar electric potential and $\vec{A}$ is the 3-D magnetic vector potential.

We know that charge is locally conserved, which is expressed by the continuity equation:

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \tag{52}$$

For a fixed reference frame,

$$\nabla \cdot \vec{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = \sum_{i=1}^{3} \frac{\partial J^i}{\partial x^i}$$

and

$$\frac{\partial \rho}{\partial t} = \frac{\partial (c\rho)}{\partial (ct)} = \frac{\partial J^0}{\partial x^0} \tag{53}$$

which leads us to conclude (using 19),

$$\frac{\partial J^\mu}{\partial x^\mu} = 0 \tag{54}$$

This is where the notation really shines - it places time and space on an equal footing and tells us something deeper about this object J^μ that we've formed. Precisely, that it is divergence-less in 4D space-time.

Encouraged by this delightful expression, we will now express Maxwell's equations as 2 very elegant equations. And the objects that will show up - you guessed it - are the field tensor and the dual tensor.

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = \mu_0 J^\mu \tag{55}$$

$$\frac{\partial G^{\mu\nu}}{\partial x^\nu} = 0 \tag{56}$$

Choosing specific values for μ in the first equation would give rise to Gauss' law ($\mu = 0$) and Ampère's law with Maxwell's correction ($\mu = 1, 2, 3$ give the 3 vectorial components of the eq.). Doing the same for the second equation would give us the eq. exhibiting the

[6] It's the Hodge dual of $F^{\mu\nu}$

divergence-less nature of B-fields and Faraday's law.

We can also express the Minkowski force on a charge K^μ using the field tensor as follows,

$$K^\mu = q\eta^\nu F^{\mu\nu} \tag{57}$$

Evaluating the zeroth component of this equation gives us (with the three dimensional dot product),

$$K^0 = \frac{\gamma q}{c}(\vec{u} \cdot \vec{E}) = \frac{1}{c}(\vec{F}_E \cdot \vec{\eta}) \tag{58}$$

The field tensor can be expressed in terms of the 4-vector potential.

$$F^{\mu\nu} = \frac{\partial A^\mu}{\partial x_\nu} - \frac{\partial A^\nu}{\partial x_\mu} \tag{59}$$

Note that the derivatives now carry a subscript instead of superscript, so this would cost us a minus sign for the temporal component (w.r.t. our original convention).

5 Density Matrices

6 Generators of transformations

7 Transverse and Longitudinal components of fields

8 Plasmas

9 Retarded potentials and Maxwell's stress tensor

10 Contravariance and Covariance

11 Dispersion relations

12 Redshifts and time in Cosmology

13 Classical Field Theory

References